



University of Asia Pacific

CSE 302: Object Oriented Programming II Lab: Visual and Web Design
Project Documentation

BookLoop Library

Submitted to
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Submitted by

Group Name: Status 200 OK

Group Number: 01

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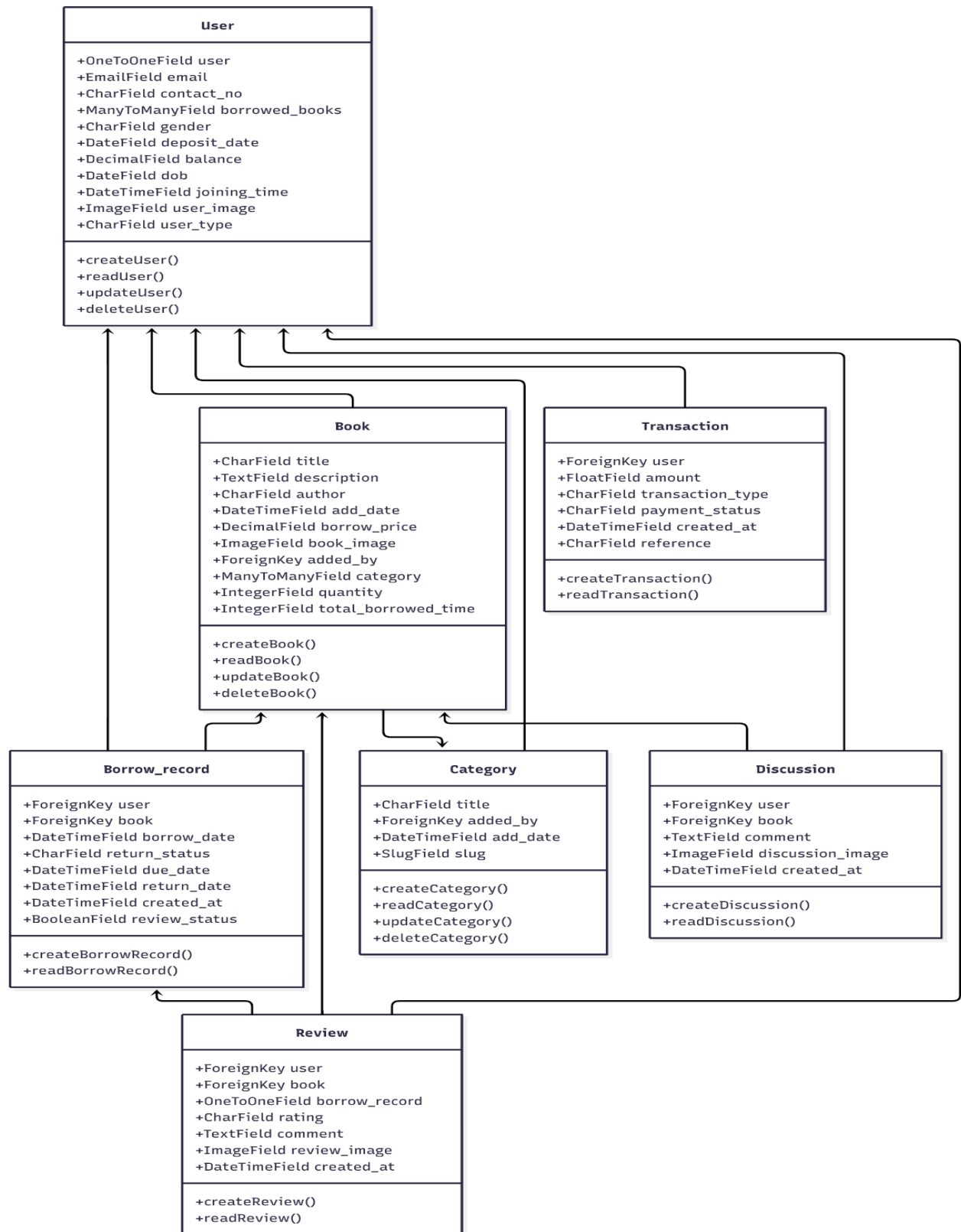
Introduction:**Description:**

BookLoop is a responsive web-based library management system built with Django, Jinja Templates, Tailwind CSS, and DaisyUI. It supports three user roles—Reader, Administrator, and Superuser. Readers can borrow books, review them, and view their transaction and borrowing history. Administrators can additionally manage books and categories, while the Superuser can promote readers to administrators.

Motivation:

The project aims to create a hassle-free, cashless, and open-source library system that simplifies book borrowing, enhances transparency for both readers and authorities, and promotes knowledge sharing within a connected digital community.

Class Diagram



- 1. Book:** Represents a book in the library.
 - **FKs:** added_by → who added the book; category → related book categories.
- 2. Borrow_record:** Tracks each book borrowing activity.
 - **FKs:** user → borrower; book → borrowed book.
- 3. Category:** Defines book genres or groups.
 - **FK:** added_by → who created the category.
- 4. Discussion:** Stores user comments on books.
 - **FKs:** user → commenter; book → discussed book.
- 5. Review:** Holds user reviews and ratings.
 - **FKs:** user → reviewer; book → reviewed book; borrow_record → related borrow instance.
- 6. Transaction:** Logs user payments and deposits.
 - **FK:** user → who made the transaction.
- 7. User:** Extends Django's base user with library info.
 - **FKs:** user → base Django user; borrowed_books → books currently borrowed.

Project Structure:

1 . Books App:

• Purpose:	
• Models:	
• Book	
	➤ title
	➤ description
	➤ author

- add_date
- borrow_price
- book_image
- added_by
- category
- quantity
- total_borrowed_time

2. Borrow_records App:

- Purpose:

- Models:

- Borrow_record

- user
- book
- borrow_date
- return_status
- due_date
- return_date
- created_at

3. Categories App:

- Purpose:

- Models:

- category

4. Discussions App:

- Purpose:

- Models:
 - Discussion

5. Reviews App:

- Purpose:
- Models:
 - Review

6. Transactions App:

- Purpose:
- Models:
 - Transaction

7. Users App:

- Purpose:
- Models:
 - User

Installation and Setup:

- 1. Clone the Repository:** First, clone the project repository from GitHub:

```
git clone <repository_url> && cd BookLoop
```

- 2. Set up Python Environment:** Ensure that Python (version 3.8 or higher) is installed. Then, create and activate a virtual environment:

```
python3 -m venv .venv
```

```
source .venv/bin/activate    # For Linux/Mac
```

```
.venv\Scripts\Activate.ps1  # For Windows
```

- 3. Configure Payment Gateway:** Create a Sandbox Account on SSLCommerz Obtain your sandbox API credentials. Create Environment File Copy the sample environment file and rename it to .env:

```
cp .env_sample .env
```

Then, fill in the necessary variables such as SECRET_KEY, Store_ID, Store_Password, Issandbox, SSLZ_URL and SSLCommerz sandbox credentials.

- 4. Apply Database Migrations and Create Superuser:**

```
python manage.py migrate && python manage.py  
createsuperuser
```

- 5. Run the Development Server:**

```
python manage.py runserver 0.0.0.0:8000
```

The project will now be accessible at <http://localhost:8000>

Contribution

- Public github link—
<https://github.com/Arannamoy-Mondal/BookLoop>
- Screenshot of the of contribution of each member:



Table 1: Contribution of each member

SL	Full Name	Reg ID	Github username	Number of Commits	No of Lines added	No of Lines deleted
1.	Nafis Fuyad Niloy	22201223	NFNiloy	8	661	103
2.	Md. Jubaer	23101008	rafnanjubaer	9	591	183
3.	Md. Osman Goni	23101012	Water-Pot	15	1158	661
4.	Aeysha Tabassum	23101014	Aeysha14	9	590	160
5.	Arannamoy Mondal	23101024	Arannamoy-Mondal	66	8926	6481

Project Summary Table

Components	Required	This project	Summery
Number of Apps	3	7	Name of the apps: users, books, categories, borrow_records, transactions, discussions, reviews
Number of Classes	7	7	Name of the classes: User, Book, Category, Borrow_record, Transaction, Discussion, Review
Number of foreign keys	2	11	Foreign keys are used in the following 6 classes, and they are - Book, Category, Borrow_record, Transaction, Discussion, Review
Use of Media	2	4	Media is used in the following 4 classes, and they are - User, Book, Discussion, Review

CEP Justification Table

Aspect	Attribute	Justification for Django E-commerce Project	CLO	PLO
Knowledge (K)	K2 – Mathematics, numerical analysis, information science	Required for fine-tuning database queries, calculating book borrowing prices, and handling user transactions accurately.	CLO1	PLO1
	K3 – Engineering fundamentals	Understanding web architecture, database normalization, and data relationships between users, books, and transactions.	CLO1	PLO1
	K4 – Specialist knowledge	Application of Django ORM, secure SSL payment gateway integration, and relational database design for a book borrowing system.	CLO1, CLO3	PLO1, PLO5
	K6 – Knowledge of practice	Deployment on cloud (e.g., Firebase/Netlify), managing APIs, and maintaining system updates through CI/CD practices.	CLO2, CLO3	PLO3, PLO5
	K7 – Role of engineering in society	Promotes digital literacy, fair access to educational resources, and ensures ethical data handling for users.	CLO4	PLO6
Problem (P)	P1 (mandatory): Requires in-depth engineering knowledge	Involves integration of user authentication, transactions, category mapping, and database relations using multiple knowledge areas (K2–K7).	CLO1	PLO1
	P2: Conflicting requirements	Trade-offs between fast data access and secure payment/storage; optimizing book search speed vs. server cost.	CLO2	PLO2
	P3: No obvious solution	Designing dynamic borrow/return flow, handling overdue penalties, and preventing double borrowing requires creative logic.	CLO2, CLO7	PLO2, PLO3, PLO10

	P6: Multiple stakeholders	Students, librarians, administrators, and payment gateway providers interact through different modules.	CLO4, CLO6	PLO6, PLO9
	P7: Interdependent parts	Book, Borrow_record, Review, Discussion, and Transaction systems are tightly linked and depend on one another.	CLO2	PLO3
Activities (A)	A1: Diverse resources	Requires cloud hosting, database servers, the Django framework, SSL payment API, and admin/user collaboration.	CLO2, CLO6	PLO3, PLO9
	A2: Conflicting issues	Balancing data security, payment accuracy, and quick response for borrowing and returning operations.	CLO2, CLO4	PLO2, PLO6
	A3: Creative use of knowledge	Applying analytics to track borrowing trends, predict popular books, and enhance user experience with recommendations.	CLO2, CLO7	PLO3, PLO10

Screenshot

BookLoop [Browse Books](#) [Login](#)

Create your Account

USERNAME
 FIRST NAME
 EMAIL ADDRESS
 CONTACT NO
 GENDER
 PASSWORD PASSWORD CONFIRMATION
[Sign Up](#)

October 2025 ↑ ↓

S	M	T	W	T	F	S
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Clear Today

mm/dd/yyyy

Sign Up



 Browse Books

 Login

Log in to your account

Or [sign up for a new account](#)

Username

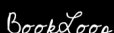
Password

Log In

Copyright © 2025 - All right reserved by Status 200 Ok



Login



 Browse Books

 Borrow History

 My Review

 Profile

 Deposit

 Transaction History

 1000.00 BDT

 Logout

Please deposit first.

Amount

Deposit

Deposit



 Browse Books

 Borrow History

 My Review

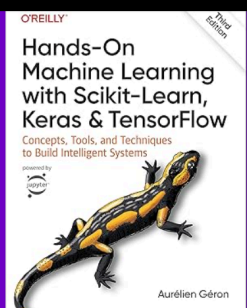
 Profile

 Deposit

 Transaction History

 6000.00 BDT

 Logout



Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems

Author: Aurélien Géron

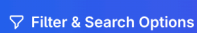
Handpicked books for you.

Total Borrowed Time: 1

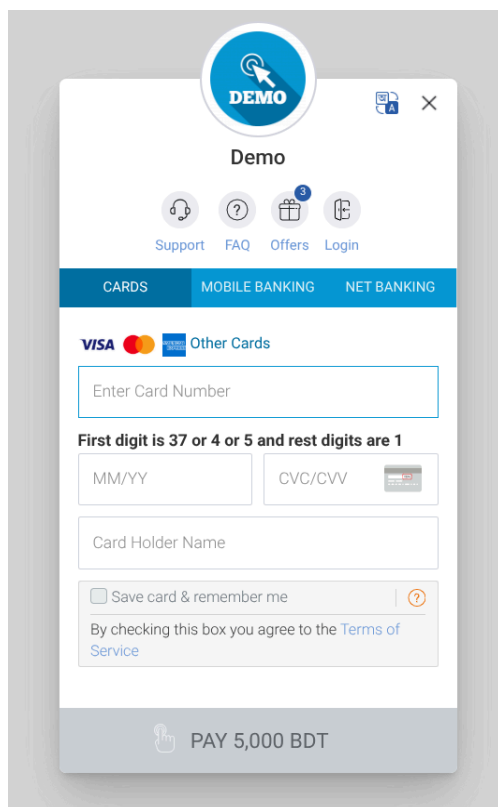


Welcome to BookLoop

Discover, borrow, and enjoy thousands of books



Homepage



DEMO

Support FAQ Offers Login

CARDS MOBILE BANKING NET BANKING

VISA MasterCard Other Cards

Enter Card Number

First digit is 37 or 4 or 5 and rest digits are 1

MM/YY CVC/CVV

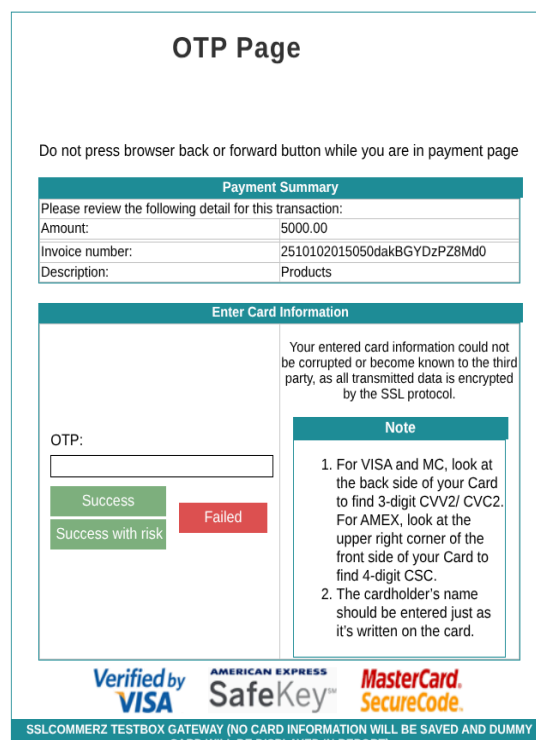
Card Holder Name

☐ Save card & remember me

By checking this box you agree to the [Terms of Service](#)

PAY 5,000 BDT

Payment Gateway



OTP Page

Do not press browser back or forward button while you are in payment page

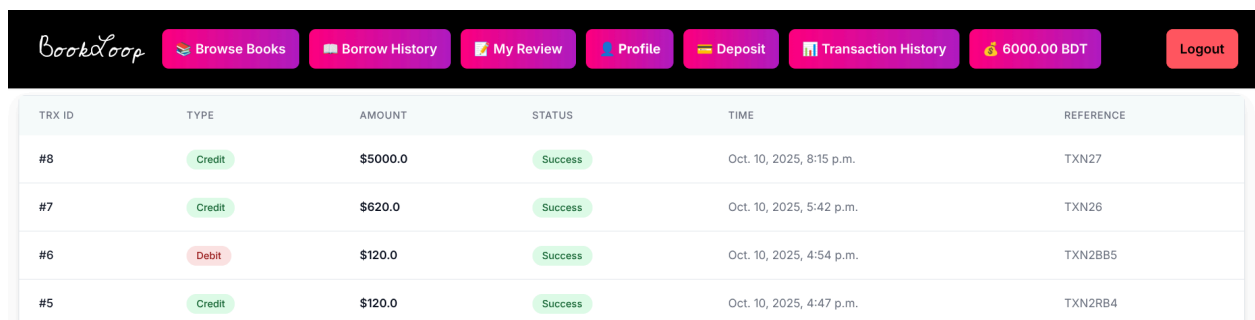
Payment Summary	
Please review the following detail for this transaction:	
Amount:	5000.00
Invoice number:	2510102015050dakBGYZPZ8Md0
Description:	Products

Enter Card Information	
OTP: Success Success with risk Failed	Your entered card information could not be corrupted or become known to the third party, as all transmitted data is encrypted by the SSL protocol. Note - For VISA and MC, look at the back side of your Card to find 3-digit CVV2/ CVC2. For AMEX, look at the upper right corner of the front side of your Card to find 4-digit CSC. - The cardholder's name should be entered just as it's written on the card.

Verified by VISA AMERICAN EXPRESS SafeKey MasterCard SecureCode.

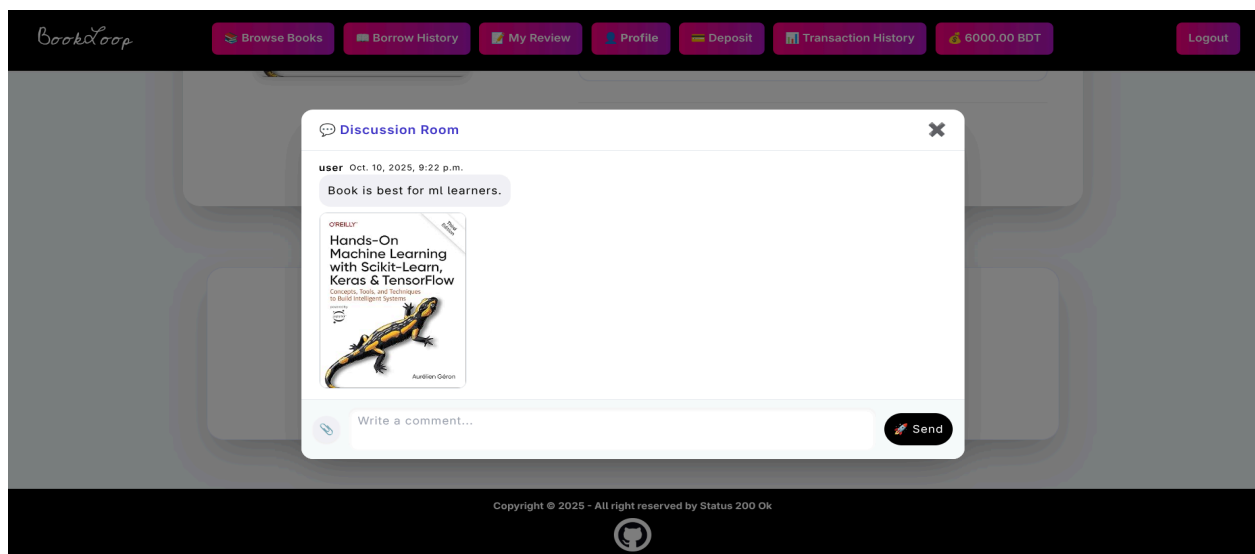
SSLCOMMERZ TESTBOX GATEWAY (NO CARD INFORMATION WILL BE SAVED AND DUMMY CARD WILL BE DISPLAYED IN RESPONSE)

Otp verification



TRX ID	TYPE	AMOUNT	STATUS	TIME	REFERENCE
#8	Credit	\$5000.0	Success	Oct. 10, 2025, 8:15 p.m.	TXN27
#7	Credit	\$620.0	Success	Oct. 10, 2025, 5:42 p.m.	TXN26
#6	Debit	\$120.0	Success	Oct. 10, 2025, 4:54 p.m.	TXN2BB5
#5	Credit	\$120.0	Success	Oct. 10, 2025, 4:47 p.m.	TXN2RB4

Transaction History



Discussion Room

user Oct. 10, 2025, 9:22 p.m.

Book is best for ml learners.

O'REILLY
Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow
Concepts, Tools, and Techniques to Build Intelligent Systems
Aurélien Géron

Write a comment...

Send

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Discussion Room On A Specific Book

BookLoop

Browse Books

Borrow History

My Review

Profile

Deposit

Transaction History

6000.00 BDT

Logout

My Borrow

Track your borrowed

Total Borrowed

4

Borrow History

BORROW ID	BOOK DETAILS	Due Date	Return Date	Action
3	THE ART OF BEING ALONE	7 days	Oct 10, 2025 8:16 PM	Return Now
4	CONCEPTS, TOOLS, AND TECHNIQUES TO BUILD INTELLIGENT SYSTEMS	7 days	Oct 17, 2025 8:16 PM	Return Now

Write a Review

Your Rating:

★ ★ ★ ★ ★

Your Comment:

Good

Attach Image:

Choose File

Submit Review

Write Review

BookLoop

Browse Books

Borrow History

My Review

Profile

Deposit

Transaction History

6000.00 BDT

Logout

👤

user one

user@user.com

📞 0123456789124

♂ Male

🎂 Aug. 12, 2002

Update Profile

First Name

user

Last Name

one

Contact No

0123456789124

Profile Image

CHOOSE FILE

No file chosen

Update Profile

